Math 3
VII-2 Solve Equations Algebraically: Infinite/No Solutions

Name:_____
Date: _____ Per: ___

Imagine a balanced scale with 3 bags and 2 cubes on one side, and 3 bags and 2 cubes on the other. Model this:



Describe the steps you would take to determine how many cubes each bag contains.

Imagine a balanced scale with 1 bag and 5 cubes on one side, and 1 bag and 2 cubes on the other side.



Describe the steps you would take to determine how many cubes each bag contains.

Equations with No Solution/Infinite Solution/One Solution

One Solution: Solve and get x = #

No Solution: Solve, variables are eliminated, false statement remains

Infinite Solutions: Solve, variables eliminated, true statement remains.

Notes:

1) $8 + 5(x - 2) = 2x - 2 + 3x$

2) $8x - 4 - 2x = 3(2x - 1)$

3) $2(3x + 1) = 3x + 1 + 3x$

4) $6x + 10 = 10 - 2x + 10x$

Turn Over

You Try/Practice:

1) $3(5x + 2) = 15x$

2) $-2(4x + 1) = -8x - 2$

3) $4(1 - x) = -4x + 4$

4) $6x - x = \frac{5}{6}(6x - 60)$

5) $10x + 7 = -3 - 10x$

6) $3(2x - 2) = 2(3x - 3)$

7) $5(x + 7) = 12x + 35$

8) $7 + 3x - 12x = 3x + 1$

9) $6x + 8 = 14 + 3x$

10) $5(1 + 2x) = \frac{1}{2}(8 + 20x)$

For items 11 & 12, write a number in the blank to create an equation with the given number of solutions.

11) -3(2 - 4x) = -6 + ___x Infinite Solutions

12) -4x + 12 - 3x = ___x + 9 No Solution